Real-Time Detection of Halo Coronal Mass Ejections from Solar Wind Data

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***Abstract*—Coronal Mass Ejections (CMEs) are large-scale eruptions of magnetized solar plasma that, when Earth-directed, disrupt the power, positioning, timing and communication systems on which modern critical infrastructure depends. Conventional CME detection relies on coronagraph image analysis, such as the CACTUS-LASCO catalogue, which is inherently retrospective and unsuitable for onboard or near-real-time alerting. This work presents a physics-informed machine learning framework that detects Halo (Type-IV) CME events directly from in-situ solar wind plasma parameters measured by the Solar Wind Ion Spectrometer (SWIS) onboard India's Aditya-L1 mission. Raw Level-2 SWIS measurements are cleaned, downsampled to a five-minute cadence, and synchronized with the CACTUS-LASCO Halo CME catalogue to construct three-day event windows. Four domain-derived composite features – the alpha-to-proton density ratio, an alpha-enrichment-over-speed-variability index, an alpha-to-thermal-pressure ratio, and the short-timescale proton-speed standard deviation – are engineered to capture the compositional and turbulence signatures that distinguish CME ejecta from ambient solar wind, and survive a screening of eight candidate physics-informed indicators. A soft-voting ensemble of Random Forest, XGBoost, and class-weighted Logistic Regression is trained on these features with a probability decision threshold of 0.45 chosen to prioritize recall, consistent with the safety-critical nature of space-weather alerting. On a held-out, stratified test split the ensemble achieves 85% accuracy, a macro F1-score of 0.84, and zero false negatives, with interval estimates reported throughout to reflect the small size of this intrinsically rare event class. The trained model is deployed as an interactive web application that accepts raw SWIS CSV uploads and returns an instantaneous CME probability score.**

***Index Terms*—coronal mass ejection, space weather, Aditya-L1, solar wind ion spectrometer, physics-informed machine learning, ensemble learning, critical infrastructure protection, real-time detection**

I. Introduction

A. Background

Coronal Mass Ejections (CMEs) are among the most energetic transient phenomena in the heliosphere, expelling billions of tonnes of magnetized plasma from the solar corona into interplanetary space. When directed toward Earth, the resulting geomagnetic disturbances can degrade GPS accuracy, induce currents in power transmission networks, disrupt high-frequency radio communication, and threaten the health of satellite electronics [1]. These consequences are overwhelmingly cyber-physical: the assets affected are the same power, positioning, timing and communication systems that underpin digital infrastructure, and the May 2024 geomagnetic storm produced exactly this pattern of coupled failures across aviation, satellite operations and GNSS-dependent services [10]. Because the source of the disturbance is external, unauthenticated and non-attributable, no access-control or perimeter mechanism applies to it, and the only viable countermeasure is advance warning with enough lead time for operators to enter a protective posture. Detection latency therefore translates directly into residual risk, which makes timely detection of CME-driven disturbances a central problem both in operational space-weather forecasting and in critical-infrastructure resilience.

India's Aditya-L1 mission, stationed at the Sun–Earth Lagrange point L1, carries the Solar Wind Ion Spectrometer (SWIS), which provides high-cadence in-situ measurements of solar wind ion parameters, including proton density, proton bulk speed, proton thermal temperature, and alpha-particle (He^{2+}) density [6]. These plasma parameters undergo characteristic changes as a CME sheath and its ejecta pass the spacecraft, offering a data-driven route to CME identification that does not depend on coronagraph imagery.

B. Motivation

The dominant CME-cataloguing approach, exemplified by the CACTUS automated recognition system applied to LASCO coronagraph images [2], is retrospective: it identifies CMEs after they have already left the corona and does not directly characterize the plasma that will later arrive at a monitoring spacecraft. In-situ, plasma-based classification could instead support near-real-time onboard alerting once a CME-

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like plasma signature is observed at the spacecraft. However, most existing in-situ studies establish empirical correlations between plasma parameters and known CME arrival times [3] rather than building an automated, generalizable classifier, while recent learning-based in-situ detectors are heavy sequence models trained on decades of multi-instrument data [5]. This leaves a gap between physically grounded solar-wind analysis and a lightweight, deployable machine learning system – a gap this paper sets out to close.

C. Objective and Contributions

This paper designs, trains, and deploys a physics-informed ensemble classifier for Halo CME detection using Aditya-L1 SWIS Level-2 data. Its main contributions are:

- An end-to-end preprocessing pipeline that converts raw SWIS CDF files into a clean, five-minute-cadence master dataset and synchronizes it with the CACTUS-LASCO Halo CME catalogue to construct labeled three-day event windows.
- A disciplined, evidence-based feature-selection process: eight physics-motivated candidate indicators are derived from first principles and screened for class separability, from which four are retained as the final, compact feature set.
- A soft-voting ensemble of Random Forest, XGBoost, and Logistic Regression classifiers, tuned with a recall-prioritizing decision threshold, that achieves zero false negatives on held-out Halo CME events despite a small and intrinsically rare positive class, with confidence intervals and an exact significance test reported alongside every point estimate.
- A deployed, interactive web application that accepts real-world SWIS CSV uploads and returns an instantaneous CME probability, demonstrating that the pipeline generalizes beyond the offline research notebook in which it was developed.

The remainder of this paper is organized as follows. Section II reviews related work in CME detection. Section III describes the dataset and preprocessing pipeline. Section IV presents the proposed feature-engineering and ensemble-modeling methodology. Section V reports experimental results and error analysis. Section VI describes the deployed system. Section VII discusses limitations, and Section VIII concludes with directions for future work.

II. Related Work

Automated CME detection has traditionally been approached through coronagraph image processing. Robbrecht and Berghmans [2] introduced the CACTUS algorithm, which automatically detects CME-like features in LASCO coronagraph sequences in near-real time; while effective for cataloguing, this approach characterizes the corona rather than the arriving plasma and offers no direct information about in-situ conditions at a monitoring spacecraft. Zhang et al. [3] established empirical correlations between in-situ solar wind parameters and major geomagnetic storms, showing that plasma-parameter excursions are informative of CME-driven disturbances, but stopped short of proposing an automated classifier. Vourlidas et al. [4] combined kinematic and compositional analysis across multiple spacecraft to characterize the physical nature of CMEs – a thorough approach, but one that is inherently post-event and multi-mission dependent rather than real-time and single-platform.

More recently, learning-based approaches have been applied on both sides of the problem. Nguyen et al. [5] applied convolutional networks over sliding windows of WIND in-situ data to detect interplanetary CMEs across 657 catalogued events, and Rüdisser et al. [8] introduced ARCANE, a framework for early ICME detection in streaming solar wind data using a deep segmentation architecture. On the imagery side, Liu et al. [12] adapt a foundation segmentation model to detect CMEs from few labeled examples. Camporeale [9] surveys the broader methodological landscape and the recurring difficulties of class imbalance and small event catalogues in space-weather machine learning – difficulties that directly shape the evaluation reported here. These approaches remain either image-based or dependent on deep sequence models and multi-instrument archives, which limits their suitability for onboard, resource-constrained deployment.

The present work differs from this body of literature in four respects: (i) it operates exclusively on in-situ plasma parameters from a single instrument (SWIS) rather than remote-sensing imagery; (ii) it uses a compact set of physically interpretable composite features rather than raw or black-box embeddings; (iii) it employs a lightweight, soft-voting ensemble of classical classifiers rather than a deep network, making it feasible for low-latency or resource-constrained environments; and (iv) it is packaged as a working, user-facing prediction service rather than remaining a purely offline analysis.

III. Dataset Preparation and Preprocessing

A. Data Sources

Two primary data sources were used in this study, summarized in Table I.

TABLE I
Datasets Used in This Study

Dataset	Description	Coverage
SWIS Level-2 (Aditya-L1)	In-situ proton density, proton speed, proton temperature, and alpha-particle density, in CDF format [6]	≈150 daily files, Aug–Dec 2024
CACTUS-LASCO CME Catalogue	Automatically detected CME events with timestamp, angular width, velocity, and halo type [7]	Daily logs, 2024–2025

B. Preprocessing Pipeline

Raw Level-2 SWIS CDF files were parsed using the `cdflib` library. For each file, the epoch time variable was

identified in a version-tolerant manner (SWIS Level-2 releases expose the epoch under different field names across CDF revisions, e.g. `epoch_for_cdf_mod`, `epoch_for_cdf`, or `Epoch`), and the four core plasma parameters were extracted using a matching field-mapping scheme. Fill values (-1.0×10^{31}) were replaced with `NaN`, and rows with no valid measurement across all four parameters were discarded. Files belonging to an early, deprecated CDF revision (V01) were excluded outright to avoid inconsistent field definitions, and only the superseding V02 releases were retained. The cleaned per-file records were concatenated chronologically into a single master dataset, `swis_master_cleaned.csv`.

The master dataset was then resampled to a uniform five-minute cadence using `pandas.resample`. For each of the four parameters, the mean, maximum, and standard deviation within each five-minute bin were computed, yielding a twelve-column downsampled feature table (`swis_downsampled_5min.csv`) that preserves both the central tendency and the short-timescale variability of the plasma flow – the latter being particularly relevant for detecting CME-associated turbulence.

C. Event Window Construction

The CACTUS-LASCO catalogue, distributed as pipe-delimited daily text logs, was parsed into a structured table retaining ten fields per event (identifier, timestamp, onset duration, position angle, angular width, velocity, velocity uncertainty, minimum and maximum velocity, and halo type). Only events classified as full Halo types (I–IV) were retained, and Halo Type-IV events – the strongest, most clearly Earth-directed class – were selected as positive CME instances. For each Halo-IV event timestamp t_0, a three-day window spanning $t_0 - 1$ day to $t_0 + 2$ days was extracted from the downsampled SWIS data, capturing the pre-event ambient wind, the shock/sheath transition, and the CME ejecta passage; each row in the window is additionally tagged with its signed offset in days from t_0. An analogous procedure was applied to timestamps that CACTUS classified as non-halo or left unclassified, from which a pool of 90 candidate quiet-period windows was constructed to serve as negative instances. This yielded a labeled corpus of 13 Halo-IV CME event windows and a reserve pool of 90 non-CME event windows, of which 30 were used for model training (Section IV-D).

D. On the Size of the Positive Class

It is worth stating plainly why the positive class is small, because the constraint is physical rather than procedural. Full Halo CMEs of the strongest, most clearly Earth-directed class are genuinely rare events, and the Halo-IV events used here constitute the complete set catalogued within the SWIS Level-2 coverage period rather than a subsample drawn from a larger pool. The count cannot be increased by additional processing, by relabeling, or by more aggressive windowing of the same telemetry: it is fixed by how many such eruptions the Sun produced while the instrument was observing. Two factors compound this. Aditya-L1 is a young mission, so the available Level-2 archive is short and no earlier SWIS observations exist to extend it backwards [11]; and Halo CME occurrence is modulated by the solar cycle, so the arrival rate of new positive examples is set by solar activity rather than by experimental design. We therefore treat the limited sample as a structural property of the problem and respond to it by reporting interval estimates alongside every point estimate (Section V-C), rather than by overstating the confidence that so few events can support. The corpus will grow as the mission archive lengthens, and the pipeline requires no methodological change to accommodate that growth.

IV. Proposed Methodology

A. Physics-Informed Feature Engineering

Rather than feeding raw time-series windows directly into a classifier, four physically motivated composite ratios are derived, chosen because heliophysical literature associates them with known differences between CME ejecta and ambient solar wind. Table II summarizes the retained features and their physical rationale. Each is computed at every five-minute bin and averaged over the three-day interval to yield a single scalar per window.

These four features were selected after an exploratory search over eight candidate physics-informed indicators: the alpha–proton ratio, an alpha-surge index (ratio of peak to mean alpha density), a delta-speed shock indicator (pre- versus post-event mean speed difference), a plasma-beta proxy ($T_p n_p / v_p^2$), a rolling standard deviation of proton speed, a distribution-skewness indicator, a minimum thermal-pressure-to-density ratio (and its inverse), and several second-order combinations of alpha density with speed variability (including a flux proxy and an alpha-flux proxy). Each candidate was evaluated by comparing its kernel-density distribution and boxplot across CME and non-CME windows (Fig. 1 summarizes where this screening step sits in the overall pipeline); only the four indicators exhibiting visibly separable distributions between the two classes were retained, while the remaining candidates were discarded to keep the final feature space compact and interpretable rather than over-engineered. We note in Section VII that this screening is distributional rather than algorithmic, and identify a formal importance-ranking study as required future work.

B. Ensemble Classification Model

The four engineered features are used to train a `VotingClassifier` that aggregates three complementary base learners:

- **Random Forest (RF):** an ensemble of decision trees configured with `class_weight="balanced"` to counteract the class imbalance between the 13 CME and 30 non-CME samples, and capable of modeling non-linear feature interactions.
- **XGBoost (XGB):** a gradient-boosted tree ensemble in which the positive-class scaling factor `scale_pos_weight` is set to the ratio of negative

TABLE II
PHYSICS-INFORMED FEATURES USED IN THE FINAL ENSEMBLE MODEL

Feature	Formula	Physical Rationale
Alpha–Proton Ratio (α/p)	n_α/n_p	CME ejecta is typically enriched in alpha particles relative to the ambient solar wind, so an elevated α/p ratio is a classic compositional signature of a CME core.
Proton-Speed Variability (σ_{v_p}, `vp_std_15min`)	centered rolling $\mathrm{std}(v_p)$ over three consecutive 5-minute bins (a 15-minute window)	CME ejecta tends to flow in a smooth, magnetically ordered manner, producing *lower* short-timescale speed variability than the turbulent ambient wind or sheath region.
Alpha over Speed-Variability Index (`alpha_over_vpstd`)	$\dfrac{(n_\alpha/n_p)}{\sigma_{v_p}}$	A composite indicator combining alpha enrichment with flow smoothness; large values reflect the co-occurrence of alpha-rich, low-turbulence ejecta characteristic of CME cores.
Alpha-to-Thermal-Pressure Ratio (`alpha_tp_ratio`)	n_α/T_p	CME ejecta is often cooler than the ambient wind for a given density, so a high alpha density relative to proton temperature highlights compositionally distinct, thermally depleted plasma.

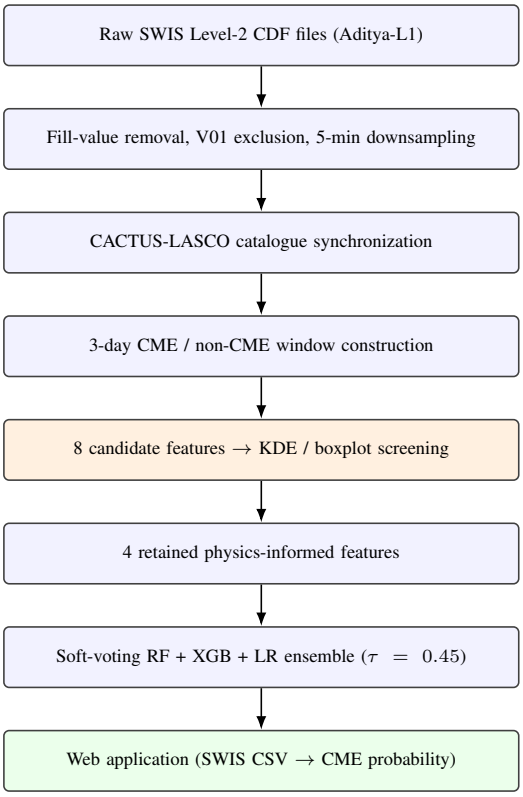


Fig. 1. End-to-end pipeline: from raw SWIS telemetry to a deployed CME probability score. The orange node denotes the feature-screening decision stage and the green node the operational output.

to positive training samples, with `logloss` as the evaluation metric.

- **Logistic Regression (LR):** an ℓ_2-regularized linear baseline with `class_weight="balanced"`, solved using the `liblinear` solver, providing a simple, well-calibrated probability estimate that complements the tree-based learners.

The three base classifiers are combined through *soft voting*: their predicted class probabilities are averaged rather than their hard class labels, which produces a smoother, better-calibrated ensemble probability estimate and tends to reduce variance relative to any single base model.

C. Recall-Oriented Threshold Tuning

Because a missed CME detection carries disproportionately high operational risk in a space-weather alerting context, the standard 0.5 probability cutoff is replaced with a tuned threshold of $\tau = 0.45$ on the ensemble's predicted CME probability:

$$\hat{y} = \begin{cases} 1 \text{ (CME)}, & P(\text{CME} \mid x) \geq \tau \\ 0 \text{ (Non-CME)}, & \text{otherwise} \end{cases} \tag{1}$$

Lowering the threshold below the default trades a modest increase in false positives for a reduction in false negatives, consistent with a design philosophy that treats missed detections as materially more costly than false alarms. In an infrastructure-protection setting this asymmetry is concrete: a precautionary posture change costs deferred maintenance and reduced transmission margin, whereas an unmitigated severe event costs transformer replacement and multi-day outage. Algorithm 1 summarizes the end-to-end training and thresholding procedure.

D. Training Procedure

The combined feature set of 43 labeled windows (13 CME, 30 non-CME) was split into training and test partitions using a stratified 70/30 split (`random_state=42` for reproducibility), preserving the class ratio in both partitions. The ensemble was fit on the training partition and evaluated on the held-out test partition (13 windows) using standard classification metrics and the confusion matrix. The final trained ensemble was serialized with `joblib` for downstream deployment.

V. EXPERIMENTS AND RESULTS

A. Feature Separability

Prior to modeling, each candidate feature's per-window average was visualized with kernel density estimates and boxplots, split by class label. The four features retained in the final model (Table II) exhibited visibly distinct distributions between CME and non-CME windows, whereas the discarded candidates showed substantial class overlap. Qualitatively,

Algorithm 1 Physics-informed ensemble training

Require: Downsampled SWIS table S, CACTUS catalogue C, threshold $\tau = 0.45$

Ensure: Trained ensemble E and test-set metric estimates

Extract Halo-IV timestamps from C; build 13 three-day CME windows from S

Sample 30 non-CME windows from the 90-window quiet-period pool

for each window w **do**

  Compute α/p, σ_{v_p}, `alpha_over_vpstd`, `alpha_tp_ratio`

end for

Assemble feature matrix $X \in \mathbb{R}^{43\times 4}$ and labels $y \in \{0,1\}^{43}$

Stratified split: $X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}} \leftarrow$ `train_test_split`$(X, y, 0.3, 42)$

Fit RF, XGB, LR on $X_{\text{train}}, y_{\text{train}}$

Build soft-voting ensemble E from {RF, XGB, LR}

$p \leftarrow E$.`predict_proba`$(X_{\text{test}})[:, 1]$

$\hat{y} \leftarrow \mathbf{1}[p \geq \tau]$ (elementwise indicator)

Evaluate $\hat{y}$ against y_{test}; serialize E with `joblib`

return E and the test-set metric estimates

CME windows showed elevated alpha-to-proton and alpha-to-thermal-pressure ratios alongside suppressed short-timescale speed variability, consistent with the expected signature of alpha-enriched, magnetically ordered ejecta embedded in a comparatively smooth flow.

B. Classification Performance

Table III reports the classification performance of the soft-voting ensemble on the held-out test partition at the tuned decision threshold of 0.45, and Table IV gives the corresponding confusion matrix.

TABLE III
TEST-SET CLASSIFICATION PERFORMANCE (THRESHOLD = 0.45)

Metric	Non-CME	CME	Overall
Precision	1.00	0.67	–
Recall	0.78	1.00	–
F1-score	0.88	0.80	0.84 (macro)
Accuracy	0.85		

TABLE IV
CONFUSION MATRIX ON THE HELD-OUT TEST SET

	Predicted Non-CME	Predicted CME
True Non-CME	7	2
True CME	0	4

The most operationally significant result is that the model produced *zero false negatives*: every true CME window in the test set was correctly flagged, at the cost of two non-CME windows being misclassified as CME. This asymmetry is a direct, intended consequence of the recall-prioritizing threshold and class-weighting scheme described in Section IV, and aligns with the safety-critical nature of CME alerting, where an over-cautious false alarm is far less costly than a missed real event.

C. Statistical Reliability of the Reported Metrics

Because the test partition contains only 13 windows, point estimates alone would overstate the precision of these results. Table V therefore accompanies each rate with a 95% Wilson score interval, which is preferred over the normal approximation because it remains well behaved at small n and at proportions of 0 and 1.

TABLE V
POINT ESTIMATES WITH 95% WILSON SCORE INTERVALS

Quantity	Count	Estimate	95% CI
Recall (CME)	4/4	1.00	[0.51, 1.00]
Precision (CME)	4/6	0.67	[0.30, 0.90]
Specificity (non-CME)	7/9	0.78	[0.45, 0.94]
Overall accuracy	11/13	0.85	[0.58, 0.96]

Three further statistics characterize the same 2×2 table. Fisher's exact test returns $p = 0.021$ (two-sided), so the association between predicted and true labels is statistically significant at the 5% level despite the small sample. Agreement measures that are robust to class imbalance give a Matthews correlation coefficient of $\mathrm{MCC} = 0.72$ and Cohen's $\kappa = 0.68$, with a balanced accuracy of 0.89. Finally, applying the rule of three to zero misses in four positive trials yields a 95% upper bound of $3/4$ on the true miss rate. The claim supported by this experiment is therefore that no misses were observed on the held-out events, not that the miss rate is identically zero; establishing the latter would require a larger event sample than the phenomenon currently affords (Section III-D).

D. Error Analysis

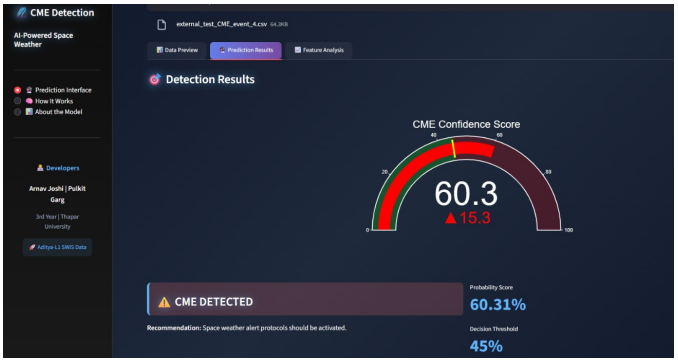
The two false positives correspond to quiet-period windows that nonetheless exhibited a transiently elevated alpha-to-proton ratio, most plausibly caused by minor stream–stream interactions or weak interplanetary shocks unrelated to a cataloged Halo CME. This is an expected failure mode rather than an incidental one: stream interaction regions compress and heat the ambient wind and can transiently mimic part of the compositional signature the model relies on, and the feature set contains no magnetic-field observable that would separate them. In an operational alerting context these are the less costly class of error, since such regions are themselves geoeffective, if more weakly than a Halo CME.

VI. SYSTEM DEPLOYMENT

To validate the operational feasibility of the proposed approach, the trained ensemble was wrapped in an interactive web application (as shown in Fig. 2) built with Streamlit and deployed as a public service. A user uploads a CSV file containing 5-minute-averaged SWIS measurements (`proton_density`, `proton_speed`, `proton_temperature`, `alpha_density`) covering



(a) Home page and CSV upload interface



(b) Prediction dashboard

Fig. 2. Deployment of the proposed Streamlit-based Halo CME detection system. (a) Home page with the CSV upload interface. (b) Prediction dashboard showing the detected CME status, confidence score, and probability estimate.

a two-to-three-day window. The application computes the same four physics-informed features used during training, feeds them through the serialized ensemble model, and returns a binary CME/non-CME prediction along with the associated probability score and a per-feature breakdown. Beyond the prediction interface, the application exposes a "How It Works" page detailing the detection pipeline and feature physics, and an "About the Model" page summarizing dataset composition, ensemble architecture, and validation metrics, making the tool usable by both technical and non-specialist operators. Feature computation and ensemble evaluation for a single three-day window complete in well under one second on a commodity CPU, with no GPU requirement and no network access to remote coronagraph imagery, so inference is not the binding constraint on alerting latency; ground-segment data availability is. This deployment demonstrates that the proposed pipeline can move from an offline research notebook to a functioning, low-latency prediction service. The deployed application is publicly accessible at https://heliopredict.streamlit.app/, allowing users to upload SWIS data and perform Halo CME prediction.

VII. Discussion and Limitations

Several limitations should be acknowledged. First, and most fundamentally, the labeled dataset is small: only the Halo-IV CME events falling within the studied observation period are available. As set out in Section III-D, this reflects the intrinsic rarity of the phenomenon and the youth of the mission archive rather than any shortcoming of the data-collection procedure, and it cannot be remedied by further processing of the same telemetry. Its consequence is that the interval estimates in Section V-C are wide and must be quoted alongside every point estimate. Second, the current model is restricted to Halo Type-IV events and does not distinguish between CME subtypes (I–III) or characterize CME intensity, so it answers "is this a strong Earth-directed event?" rather than "how severe will the resulting disturbance be?". Third, the four selected features, while physically interpretable, were chosen via visual distributional comparison rather than a formal feature-selection algorithm such as recursive feature elimination or SHAP-based importance ranking, leaving open the possibility that a more systematic search could identify additional discriminative signals; relatedly, the reported metrics come from a single stratified split rather than a cross-validated protocol, so they should be read as indicative rather than as a tight estimate of generalization error. Fourth, the feature set contains no magnetic-field observable, which is the most plausible explanation for the confusion between CME ejecta and stream interaction regions noted in the error analysis.

Finally, the present pipeline operates on pre-collected, file-based SWIS windows rather than a live telemetry stream, so the "real-time" characterization should be understood as referring to inference latency and to independence from retrospective imagery, not to a validated live-telemetry deployment. Closing that gap requires a fixed-length ring buffer holding the most recent three days of five-minute bins, updated as each new bin arrives; because all four features are simple aggregates, they admit incremental update in $O(1)$ per bin rather than $O(n)$ recomputation, so per-bin inference cost is bounded and independent of buffer length. Two further mechanisms are needed for stable operation: hysteresis, with separate raise and clear thresholds so that a single noisy bin cannot trigger a posture change downstream, and data-quality gating that abstains rather than predicts when the fraction of valid bins in the buffer falls below a threshold. Because a sliding buffer encounters a CME while only the leading portion of the ejecta has passed the spacecraft, validating that the reported recall survives such window truncation [8] is the single most important step before operational use.

VIII. Conclusion and Future Work

This paper presented a physics-informed ensemble machine learning framework for detecting Halo Coronal Mass Ejections directly from Aditya-L1 SWIS in-situ solar wind measurements, positioned as an early-warning sensor layer for the protection of critical infrastructure against a natural threat to

its availability. By combining domain-derived plasma composition and turbulence ratios with a soft-voting ensemble of Random Forest, XGBoost, and Logistic Regression classifiers, and by tuning the decision threshold to prioritize recall, the proposed system achieved 85% test-set accuracy (95% CI $[0.58, 0.96]$; Fisher exact $p = 0.021$) with no missed CME detections observed on the held-out events, despite training on only 43 labeled windows. A deployed web application further demonstrates that the model can be operated as an accessible, near-real-time decision-support tool rather than remaining confined to an offline notebook. Future work will focus on (i) extending the corpus as the Aditya-L1 archive lengthens through the current solar cycle, which is the only route by which this intrinsically rare positive class can grow, together with incorporating non-Halo CME types to sharpen the decision boundary; (ii) replacing the present distributional screening with a formal feature-importance analysis such as SHAP values, and the single stratified split with stratified k-fold cross-validation and ROC/PR-curve-based threshold selection reported against per-base-model baselines; (iii) incorporating magnetic-field observables to separate CME ejecta from stream interaction regions; and (iv) extending the classifier to a multi-class formulation capable of distinguishing CME halo types and integrating the pipeline into a continuous, streaming space-weather monitoring system.

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